

Mid Semester Examination 2025 (Set-1)

Name of the Program: B.Tech Petroleum Engineering Semester: V
Name of the Course: Petroleum Geophysics Course Code: TPE-501

Time: 1-1/2Hour

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
(a)	Explain Geophysics and its importance. What are the various earth properties measured in Geophysics to explain earth geological features?	
OR		
(b)	How Geophysical data is use in defining the subsurface structure? Draw a diagram of seismic survey.	
Q2	(10 marks)	CO 2
(a)	What are the active & passive geophysical methods of exploration? How Gravity survey explain the subsurface structure?	
OR		
(b)	What is the main objective of geophysical investigations? Explain gravity anomaly. How it is useful for structure identification?	
Q3	(10 marks)	CO 2
(a)	Explain the principle of Magnetic survey method. What kind of information Magnetic survey method will provide us? Explain in detail.	
OR		
(b)	What is the principle of Electromagnetic method for geophysical survey? Draw a neat diagram to explain it. Where this survey is useful?	
Q4	(10 marks)	CO 1 CO 2
(a)	Write the principle of seismic method for exploration. What are the basic properties measured in this method?	
OR		
(b)	Explain P-Wave and S-wave along with figure. Write the Snell's laws for P-wave and S- wave by drawing a figure.	
Q5	(10 marks)	CO 2
(a)	Explain Huygens principle for wave propagation. Draw a diagram for wave propagation for reflection and Refraction.	
OR		
(b)	What is acoustic Impedance and reflection coefficients? Write their formulas and their importance.	

